Codebook: WM-FTT working memory strategies data (v1/v2/v3), participant- and stimulus-level exports

Data from the Working Memory Feature Trade-off Task (WM-FTT): word, orientation and colour recall under a parity-judgement distractor, pooled across three task versions (v1 original, v2 added a parity-error penalty, v3 added recall-order randomisation). Two datasets are described here: all_data_surveys (one row per participant: demographics, questionnaire scores, and raw questionnaire/strategy items) and all_data_full (one row per stimulus: trial structure, targets, responses, timing, and per-feature recall scores; no questionnaire item detail, see all_data_surveys for that). NAMING NOTE: the task was called CFA-WM during data collection, and column names, OSF component names and the front-end code still use that prefix (for example strats_cfa_q01_colors_1). Those identifiers are deliberately unchanged so that this export remains byte-comparable with earlier ones and with the raw JATOS data. WM-FTT and CFA-WM refer to the same task.

Sources

- WM-FTT (formerly CFA-WM) v1/v2/v3, Mael Delem, EMC Lab, Universite Lumiere Lyon 2

Instrument references and copyright notice

- VVIQ item text: Marks, D. F. (1973). Visual imagery differences in the recall of pictures. British Journal of Psychology, 64(1), 17-24. https://doi.org/10.1111/j.2044-8295.1973.tb01322.x
- OSIVQ item text: Blazhenkova, O., & Kozhevnikov, M. (2009). The new object-spatial-verbal cognitive style model: Theory and measurement. Applied Cognitive Psychology, 23(5), 638-663. https://doi.org/10.1002/acp.1473
- NIEQ item text: Heavey, C. L., Moynihan, S. A., Brouwers, V. P., Lapping-Carr, L., Krumm, A. E., Kelsey, J. M., Turner, D. K., & Hurlburt, R. T. (2019). Measuring the frequency of inner-experience characteristics by self-report: The Nevada Inner Experience Questionnaire. Frontiers in Psychology, 9, 2615. https://doi.org/10.3389/fpsyg.2018.02615

Dataset 1: `all_data_surveys` (one row per participant)

Core variables

Variable	Description	Type	Range	Levels / notes
`id`	Participant identifier, unique within a version.	string		
`version`	WM-FTT task version.	string		`v1` = Original task version; `v2` = Added a parity-error penalty to recall scoring; `v3` = Added recall-order randomisation; independent front-end infrastructure

Variable	Description	Type	Range	Levels / notes
`language`	UI language of the experiment in the browser.	string		`en` = English; `fr` = French
`age`	Participant age in years, self-reported.	integer	18-71 years	
`gender`	Gender, as freely typed by the participant.	string		Free text, not a controlled vocabulary; values include both short codes (“f”, “m”) and longer self-descriptions in French and English (e.g. “Non-binaire”, “female/agender”). Left as-is rather than collapsed into fixed categories, since collapsing would lose information some participants specifically chose to add. No missing values in the current data.
`vviq_total_score`	Total score on the Vividness of Visual Imagery Questionnaire (VVIQ; Marks, 1973), summed across all 16 items (see vviq_q01-vviq_q16 below). Lower scores indicate weaker/absent visual imagery.	integer	16-80	Missing for 2 participants (of 116).
`vviq_group_2`	Categorical VVIQ group with 2 levels, based on standard VVIQ cutoff.	string		`aphantasia` = VVIQ <= 32 (aphantasia and hypophantasia collapsed); `typical` = VVIQ > 32 (typical and hyperphantasia collapsed)

Variable	Description	Type	Range	Levels / notes
`vviq_group_4`	Categorical VVIQ group with 4 levels, based on standard VVIQ cutoffs.	string		`aphantasia` = VVIQ = 16 (complete absence of visual imagery); `hypophantasia` = VVIQ 17-32 (reduced visual imagery); `typical` = VVIQ 33-74 (typical visual imagery); `hyperphantasia` = VVIQ 75-80 (extremely vivid visual imagery)
`nieq_mental_imagery`	Nevada Inner Experience Questionnaire (NIEQ; Heavey et al., 2019) mental imagery dimension score: mean of the frequency and proportion items for this dimension (see nieq__freq__mental__imagery / nieq__prop__mental__imagery in the nested item columns).	float	0-100	No missing values.
`nieq_inner_voice`	NIEQ inner voice/speaking dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
`nieq_emotions`	NIEQ feelings/emotions dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
`nieq_sensory_focus`	NIEQ sensory awareness dimension score (mean of frequency and proportion items).	float	0-100	No missing values.

Variable	Description	Type	Range	Levels / notes
nieq_unsymbolised	NIEQ unsymbolized thinking dimension score (mean of frequency and proportion items).	float	0-100	No missing values.
object_mean	OSIVQ (Blazhenkova & Kozhevnikov, 2009) object-imagery subscale mean.	float	1-4.64	Missing for 2 participants (same 2 as vviq_total_score).
spatial_mean	OSIVQ spatial-imagery subscale mean.	float	1-5	Missing for 2 participants (same 2 as vviq_total_score).
verbal_mean	OSIVQ verbal cognitive-style subscale mean.	float	1-5	Missing for 2 participants (same 2 as vviq_total_score).
feedback_total_score_word	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative word points shown to the participant at the end of the task, summed across trials. Derived from the front end's defective edit distance; see live_diff_word. For analysis use score_word, aggregated as the question requires.	float	6-70.5	
feedback_total_score_angle	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative orientation points shown at the end of the task. For analysis use score_angle.	float	0-64.5	

Variable	Description	Type	Range	Levels / notes
`feedback_total_score_color`	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative colour points shown at the end of the task. For analysis use score_color.	float	5-64	
`feedback_total_score`	DISPLAY ONLY, NOT FOR ANALYSIS. Combined total points shown to the participant, across all three features and all trials, including the parity penalty in v2 and v3. This is the number participants were told to maximise, so it is meaningful as an account of what they were optimising, but it is not a performance measure.	float	11.5-195.5	
`prognosis`	Self-reported aphantasia status/prognosis, where collected.	string		`no` = No; `yes` = Yes
`neuro_trouble`	Self-reported neurological condition, freely typed.	string		Free text (French), not a controlled vocabulary, e.g. "TDAH", "TSA", "depression suite a un burnout". NA throughout for v3 (not asked in that version's demographics form); collected for v1/v2 only.

Variable	Description	Type	Range	Levels / notes
`treatment`	Self-reported relevant treatment.	string		`no` = No; `yes` = Yes – NA throughout for v3 (not asked in that version's demographics form); collected for v1/v2 only.

VVIQ item-level responses (`vviq_q01-vviq_q16`)

The VVIQ has no subscales; its 16 items are grouped into 4 scenes of 4 items each, each rated 1 (perfectly clear and vivid) to 5 (no image at all). Item wording is not reproduced here for copyright reasons – see Marks (1973).

Variable	Description	Type	Range	Notes
`vviq_q01`	VVIQ item 1 of 16, from the scene "a relative or friend you often see" (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
`vviq_q02`	VVIQ item 2 of 16, from the scene "a relative or friend you often see" (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
`vviq_q03`	VVIQ item 3 of 16, from the scene "a relative or friend you often see" (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q04	VVIQ item 4 of 16, from the scene “a relative or friend you often see” (items 1-4 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q05	VVIQ item 5 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q06	VVIQ item 6 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q07	VVIQ item 7 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q08	VVIQ item 8 of 16, from the scene “the rising sun” (items 5-8 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q09	VVIQ item 9 of 16, from the scene “the front of a familiar shop” (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q10	VVIQ item 10 of 16, from the scene "the front of a familiar shop" (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q11	VVIQ item 11 of 16, from the scene "the front of a familiar shop" (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q12	VVIQ item 12 of 16, from the scene "the front of a familiar shop" (items 9-12 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q13	VVIQ item 13 of 16, from the scene "a country scene with trees, mountains, and a lake" (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q14	VVIQ item 14 of 16, from the scene "a country scene with trees, mountains, and a lake" (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
vviq_q15	VVIQ item 15 of 16, from the scene "a country scene with trees, mountains, and a lake" (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.
vviq_q16	VVIQ item 16 of 16, from the scene "a country scene with trees, mountains, and a lake" (items 13-16 share this scene). Item wording not reproduced; see Marks (1973).	integer	1-5	1 = perfectly clear and as vivid as normal vision; 5 = no image at all. Missing for 2 participants overall.

OSIVQ item-level responses

34 items rated 1 (totally disagree) to 5 (totally agree), across three scales (Object, Spatial, Verbal). Item order in the raw front-end data differs between v1/v2 and v3; the columns here are named consistently across both after unnesting by name (not position). Item wording is not reproduced here for copyright reasons – see Blazhenkova & Kozhevnikov (2009).

Variable	Description	Type	Range	Notes
osivq_q01s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q02v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q04v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q05s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q06o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q07s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q08v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q09v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q11o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q12o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
`osivq_q13o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q14s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q16v`	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q17s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q18o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q20o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q23o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
`osivq_q26o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q27s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q29o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q30s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q31s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q32s`	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
`osivq_q33o`	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q34o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q35v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q37v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q39v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q40o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q41v	OSIVQ item, Verbal scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q42s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

Variable	Description	Type	Range	Notes
osivq_q43o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q44s	OSIVQ item, Spatial scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.
osivq_q45o	OSIVQ item, Object scale. Item wording not reproduced; see Blazhenkova & Kozhevnikov (2009).	integer	1-5	1 = totally disagree; 5 = totally agree. Missing for 2 participants overall.

OSIVQ subscale sums and item-completion counts

Variable	Description	Type	Range	Notes
object_score	OSIVQ object-imagery subscale sum (raw item sum; object_mean above is its mean).	integer	14-65	Missing for 2 participants overall.
spatial_score	OSIVQ spatial-imagery subscale sum (raw item sum; spatial_mean above is its mean).	integer	11-55	Missing for 2 participants overall.
verbal_score	OSIVQ verbal cognitive-style subscale sum (raw item sum; verbal_mean above is its mean).	integer	12-45	Missing for 2 participants overall.
o_count	Number of object-scale items answered (out of 11).	integer		Diagnostic/completion count, not a score.
s_count	Number of spatial-scale items answered (out of 11).	integer		Diagnostic/completion count, not a score.

Variable	Description	Type	Range	Notes
v_count	Number of verbal-scale items answered (out of 12).	integer		Diagnostic/completion count, not a score.

NIEQ item-level responses

5 dimensions (mental imagery, inner voice/speaking, feelings/emotions, sensory awareness, unsymbolized thinking), each with 2 raw items (frequency, proportion), both 0-100 sliders. Item wording is not reproduced here for copyright reasons – see Heavey et al. (2019).

Variable	Description	Type	Range	Notes
nieq_freq_mental_imagery	NIEQ Mental imagery dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_mental_imagery	NIEQ Mental imagery dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_inner_voice	NIEQ Inner voice/speaking dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Variable	Description	Type	Range	Notes
nieq_prop_inner_voice	NIEQ Inner voice/speaking dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_emotions	NIEQ Feelings/emotions dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_emotions	NIEQ Feelings/emotions dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_sensory_focus	NIEQ Sensory awareness dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Variable	Description	Type	Range	Notes
nieq_prop_sensory_focus	NIEQ Sensory awareness dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_freq_unsymbolised	NIEQ Unsymbolized thinking dimension, frequency item (how often this type of inner experience occurs). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.
nieq_prop_unsymbolised	NIEQ Unsymbolized thinking dimension, proportion item (what proportion of inner experience this type makes up). Item wording not reproduced; see Heavey et al. (2019).	integer	0-100	0-100 slider. No missing values.

Strategy report, free text (`strats_cfa_q01-q03`)

Free-text self-reported memorisation strategies, mostly in French, collected in v1/v2 only. Not a controlled vocabulary – no level list is given, as one wouldn't meaningfully summarise free text.

Variable	Description	Type	Range	Notes
strats_cfa_q01_colors_1	Free-text (mostly French) self-reported strategy for remembering colours, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only. See strats_cfa_q04_scoring_strat_* below for v3's coded strategy report.

Variable	Description	Type	Range	Notes
strats_cfa_q01_colors_2	Free-text self-reported strategy for remembering colours, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q01_colors_3	Free-text self-reported strategy for remembering colours, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q02_orientations_1	Free-text self-reported strategy for remembering orientations, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only.
strats_cfa_q02_orientations_2	Free-text self-reported strategy for remembering orientations, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q02_orientations_3	Free-text self-reported strategy for remembering orientations, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).
strats_cfa_q03_words_1	Free-text self-reported strategy for remembering words, first mention.	string		Free text, not a controlled vocabulary - collected in v1/v2 only.
strats_cfa_q03_words_2	Free-text self-reported strategy for remembering words, second mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).

Variable	Description	Type	Range	Notes
`strats_cfa_q03_words_3`	Free-text self-reported strategy for remembering words, third mention (optional follow-up).	string		Free text. Collected in v1/v2 only. Mostly missing (optional field).

Strategy report, coded (`strats_cfa_q04_scoring_strat_*`)

Controlled vocabulary, collected primarily in v3 (`strats_cfa_q04_scoring_strat_1/_2`); v1/v2 has an equivalent field with a different naming convention (`_1_1/_2_1/_3_1`).

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_scoring_strat_1	Coded self-reported memorisation strategy, first choice.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v3 only (as strats_cfa_q04_scoring_strat_1; the _1_1 column below is a v1/v2 field with a different naming convention, not the same question re-asked).

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_scoring_strat_2	Coded self-reported memorisation strategy, second choice.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v3 only.

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_scoring_strat_1_1	Coded self-reported memorisation strategy, v1/v2's equivalent first-choice field.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v1/v2 only (NA for all v3 rows).

Variable	Description	Type	Range	Levels / notes
strats_cfa_q04_scoring_strat_2_1	Coded self-reported memorisation strategy, v1/v2's equivalent second-choice field.	string		repetition = Repeated the stimulus (aloud or mentally); semantic = Formed a semantic association or link; none = No particular strategy used; spatial_body = Used body movement/position as a memory aid; spatial_cardinal = Used a cardinal/directional spatial strategy; words = Focused on the word component; colours = Focused on the colour component; orientations = Focused on the orientation component; mental_image = Formed a mental image – Collected in v1/v2 only (NA for all v3 rows). Sparsely populated even within v1/v2 (optional field).

Variable	Description	Type	Range	Levels / notes
`strats_cfa_q04_scoring_strat_3_1`	Coded self-reported memorisation strategy, v1/v2's equivalent third-choice field.	string		`repetition` = Repeated the stimulus (aloud or mentally); `semantic` = Formed a semantic association or link; `none` = No particular strategy used; `spatial_body` = Used body movement/position as a memory aid; `spatial_cardinal` = Used a cardinal/directional spatial strategy; `words` = Focused on the word component; `colours` = Focused on the colour component; `orientations` = Focused on the orientation component; `mental_image` = Formed a mental image – Collected in v1/v2 only; v3 has no third slot for this field (NA for all v3 rows - this is a real structural difference, not missing data). Rarely populated even within v1/v2: non-missing for only 7 of 88 v1 participants and 0 of 9 v2 participants.

Additional demographics

Variable	Description	Type	Range	Notes
`country`	Self-reported country of residence (free text/country code, not standardised).	string		
`job`	Self-reported occupation, using the ESEG occupational classification codes where applicable.	string		
`education`	Self-reported education level, using ISCED classification codes where applicable.	string		
`field`	Self-reported field of study or work, free text.	string		
`where_from`	Self-reported source of how the participant heard about the study.	string		
`language_native`	Participant's self-reported native language.	string		
`language_usual`	Participant's self-reported language of everyday use, where different from native language.	string		

Dataset 2: `all_data_full` (one row per stimulus)

id, version, language, age, gender, vviq_total_score, vviq_group_2, vviq_group_4, nieq_mental_imagery, nieq_inner_voice, nieq_emotions, nieq_sensory_focus, nieq_unsymbolised, object_mean, spatial_mean, and verbal_mean are the same participant-level columns described in all_data_surveys above, repeated on every stimulus row for this participant. See all_data_surveys for their descriptions, ranges, and missingness.

Variable	Description	Type	Range	Levels / notes
`expe_phase`	Task phase this stimulus belongs to.	string		`tutorial` = Single practice trial with feedback; `training` = Practice trials, no feedback; `expe_block_1` = Test block 1; `expe_block_2` = Test block 2; `expe_block_3` = Test block 3
`trial_number`	Trial index (1 tutorial + 3 training + 21 test trials per participant).	integer		
`response_order`	Order in which the three response modalities (word/angle/colour) were collected for this trial.	string		`word_angle_color` = Word, then angle, then colour; `word_color_angle` = Word, then colour, then angle; `angle_word_color` = Angle, then word, then colour; `angle_color_word` = Angle, then colour, then word; `color_word_angle` = Colour, then word, then angle; `color_angle_word` = Colour, then angle, then word – Fixed at word_angle_color in v1/v2 (always that order); randomised per trial in v3, so all 6 permutations appear.
`item_number`	Stimulus index within the trial (1-3: word, orientation, colour).	integer		
`target_word`	The to-be-remembered word for this stimulus.	string		

Variable	Description	Type	Range	Levels / notes
target_angle	The to-be-remembered rectangle orientation, in degrees.	integer		
target_color_angle	The to-be-remembered colour, as an angle on the colour wheel used for continuous colour report.	float		
target_color	The to-be-remembered colour, as its nearest named colour.	string		
response_word	Participant's recalled word.	string		
response_angle	Participant's recalled rectangle orientation, in degrees.	float		
response_color_angle	Participant's recalled colour, as an angle on the colour wheel.	integer		
response_color	Participant's recalled colour, as its nearest named colour.	string		

Variable	Description	Type	Range	Levels / notes
`score_word`	Word recall similarity, 1 = exact match. Edit-distance scoring (Gonthier, 2022): Damerau-Levenshtein distance capped at the target's length, subtracted from it, divided by it. Computed in R from the raw target and response, not from the task's own feedback columns. WARNING: word recall is near ceiling in this task and its split-half reliability is 0.45, below the conventional floor; it does not support individual-differences claims. See the Scoring page of the package documentation.	float	0-1	
`score_angle`	Orientation recall similarity, 1 = exact match. Cosine of the angular error on a 180-degree period, since a rectangle's tilt is symmetric under 180-degree rotation. Split-half reliability 0.82.	float	0-1	
`score_color`	Colour recall similarity, 1 = exact match. Cosine of the angular error on the 360-degree hue wheel. Split-half reliability 0.83.	float	0-1	

Variable	Description	Type	Range	Levels / notes
score_word_z	score_word, z-scored within task version using experimental-block rows only. Retained alongside the raw score: raw for absolute performance, standardised for cross-feature comparison.	float		
score_angle_z	score_angle, z-scored within task version. See score_word_z.	float		
score_color_z	score_color, z-scored within task version. See score_word_z.	float		
responded_word	Whether the participant entered a word at all. IMPORTANT: the task encodes non-response with sentinels rather than missing values, and non-responses are scored 0 in score_word. Analyses of recall accuracy should filter on this flag; analyses that leave it out are combining accuracy with willingness to respond, which are different quantities. See the Scoring page of the package documentation.	boolean		

Variable	Description	Type	Range	Levels / notes
`responded_angle`	Whether the participant moved the orientation widget. NOTE: unlike the other two features this is an inference, not an observation. The widget starts at 90 degrees and returns that value if untouched, so a deliberate 90-degree response is indistinguishable from no response. No block target lies within 20 degrees of 90, so such a response can never be near-correct. See responded_word.	boolean		
`responded_color`	Whether the participant moved the colour wheel. Non-response is marked in the raw data by response_color_angle == 999 and response_color '#AAAAAA'. See responded_word.	boolean		

Variable	Description	Type	Range	Levels / notes
`live_diff_word`	DISPLAY ONLY, NOT FOR ANALYSIS. Word recall dissimilarity as computed by the task's front end to drive its feedback screen, 0 = exact match. Note this runs opposite to score_word. WARNING: the front end's edit distance is defective (a loop whose condition is an assignment leaves the distance matrix half-initialised), so this is not a valid edit distance. Retained for provenance only. See the Scoring page.	float		
`live_diff_angle`	DISPLAY ONLY, NOT FOR ANALYSIS. Absolute orientation error divided by 180, as computed by the front end, 0 = exact match. Linear rather than cosine, and with no handling of the rectangle's 180-degree symmetry. Use score_angle instead.	float		

Variable	Description	Type	Range	Levels / notes
`live_diff_color`	DISPLAY ONLY, NOT FOR ANALYSIS. Circular colour error divided by 180, as computed by the front end, 0 = exact match. Non-responses are mapped to 1 by an explicit guard. Use score_color instead.	float		
`rt_word`	Response time for the word recall response.	integer		
`rt_angle`	Response time for the orientation recall response.	float		
`rt_color`	Response time for the colour recall response.	float		
`feedback_score_word`	DISPLAY ONLY, NOT FOR ANALYSIS. Points shown on the feedback screen for the word response (0, 0.5 or 1), thresholded from live_diff_word and therefore inheriting its defective edit distance. Use score_word.	float		
`feedback_score_angle`	DISPLAY ONLY, NOT FOR ANALYSIS. Points shown for the orientation response (0, 0.5 or 1). Use score_angle.	float		
`feedback_score_color`	DISPLAY ONLY, NOT FOR ANALYSIS. Points shown for the colour response (0, 0.5 or 1). Use score_color.	float		

Variable	Description	Type	Range	Levels / notes
`feedback_trial_score`	DISPLAY ONLY, NOT FOR ANALYSIS. Points shown for the whole trial: the sum of the three per-feature feedback scores, minus 0.5 per incorrect parity judgement, floored at 0. Trial-level, so constant across the three stimulus rows of a trial. The parity penalty applies in v2 and v3 only.	float		
`parity_1_stim`	First of the two parity-judgement distractor digits embedded in this trial.	integer		
`parity_1_resp`	Participant's parity judgement (odd/even) for parity_1_stim.	string		
`parity_1_acc`	Accuracy of the first parity judgement (1 = correct, 0 = incorrect).	integer		
`parity_1_rt`	Response time for the first parity judgement.	integer		
`parity_2_stim`	Second of the two parity-judgement distractor digits embedded in this trial.	integer		
`parity_2_resp`	Participant's parity judgement (odd/even) for parity_2_stim.	string		
`parity_2_acc`	Accuracy of the second parity judgement (1 = correct, 0 = incorrect).	integer		
`parity_2_rt`	Response time for the second parity judgement.	integer		

Variable	Description	Type	Range	Levels / notes
`feedback_total_score_word`	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative word points shown to the participant at the end of the task. Derived from the front end's defective edit distance; see live_diff_word. Use score_word.	float		
`feedback_total_score_angle`	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative orientation points shown at the end of the task. Use score_angle.	float		
`feedback_total_score_color`	DISPLAY ONLY, NOT FOR ANALYSIS. Cumulative colour points shown at the end of the task. Use score_color.	float		
`feedback_total_score`	DISPLAY ONLY, NOT FOR ANALYSIS. Combined total points shown to the participant, including the parity penalty in v2 and v3. This is the number participants were told to maximise, so it is meaningful as an account of what they were optimising, but it is not a performance measure.	float		
`prognosis`	Self-reported aphantasia status/prognosis, where collected.	string		`no` = No; `yes` = Yes

Variable	Description	Type	Range	Levels / notes
`neuro_trouble`	Self-reported neurological condition, freely typed.	string		Free text (French), not a controlled vocabulary. NA throughout for v3.
`treatment`	Self-reported relevant treatment.	string		`no` = No; `yes` = Yes – NA throughout for v3.